

A Synthetic Study of Stem Respiration in a Temperate Pine Stand

EcoEvidence Demo Dataset - This document is synthetic and contains no real research data.

Abstract

We created a small observational dataset to test evidence retrieval from ecological literature. Stem respiration was measured in Scots pine trees at a temperate forest site. The document reports species, sample size, site coordinates, tree diameter, measurement temperature, method, and a numeric respiration value so each field can be traced to page-level evidence.

Methods

The synthetic study was conducted at the Demo Forest Research Station (50.93 N, 13.57 E). We selected 12 *Pinus sylvestris* trees. Mean diameter at breast height (DBH) was 24.6 cm. Measurements were made at a stem temperature of 18 degrees C during July 2025.

Stem CO2 efflux was measured with a closed dynamic chamber connected to an infrared gas analyzer (IRGA). Chamber observations were used directly; no model-predicted respiration values were included.

Results

The observed mean stem respiration rate was 2.40 micromol CO2 m-2 s-1 at 18 degrees C (standard deviation 0.31 micromol CO2 m-2 s-1; n = 12 trees).

Table 1. Synthetic observations used in the public demo.

Field	Synthetic value	Evidence section
Species	<i>Pinus sylvestris</i>	Methods
Sample size	12 trees	Methods / Results
DBH	24.6 cm	Methods
Stem respiration	2.40 micromol CO2 m-2 s-1	Results

Limitations

All values in this PDF are invented for software testing. They must not be cited as scientific evidence.